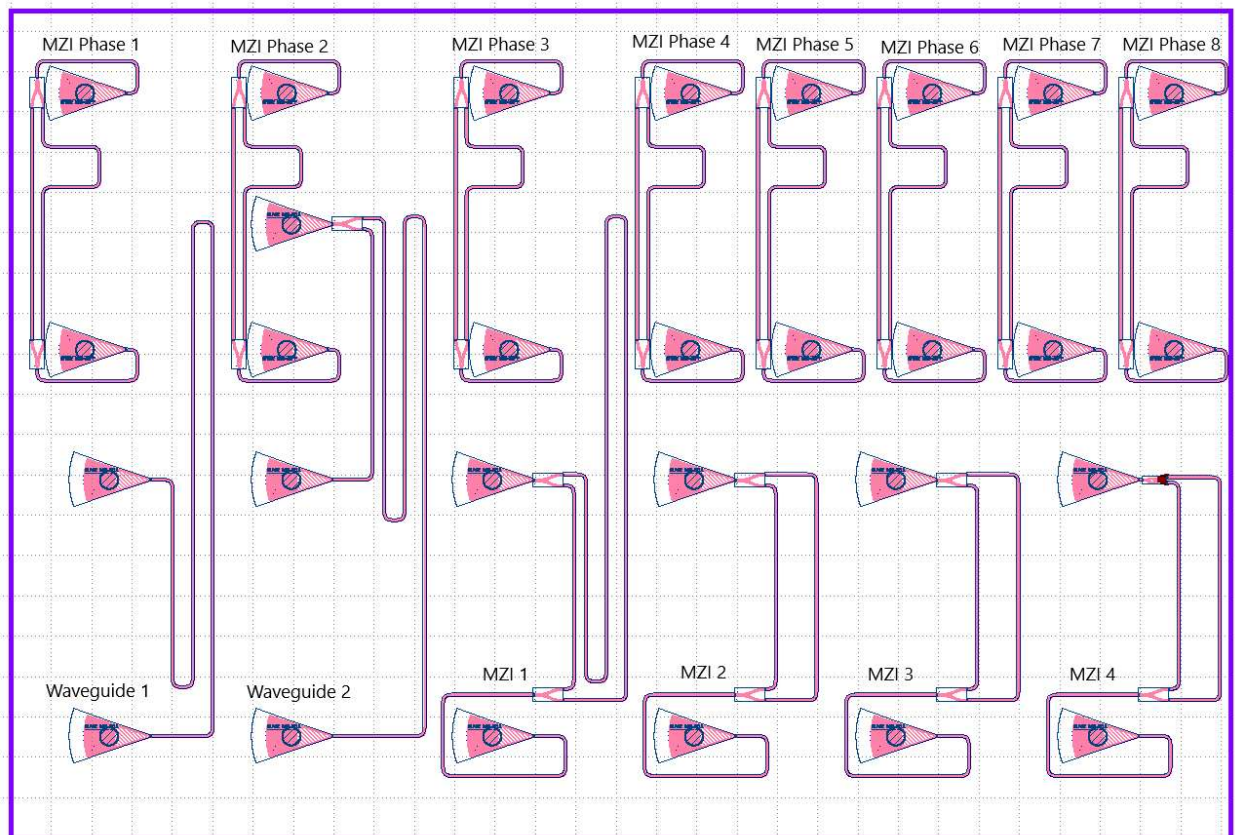


EBeam_ChrisHamner.gds design information

My purpose for taking this course is to familiarize myself with the Open_EBL process for future use research use. My layout consists of two test waveguides, four test MZI structures, and six phase shifted MZI structures. The system uses 500nm wide waveguides and is designed for TE polarization. Their purpose was chosen as follows:



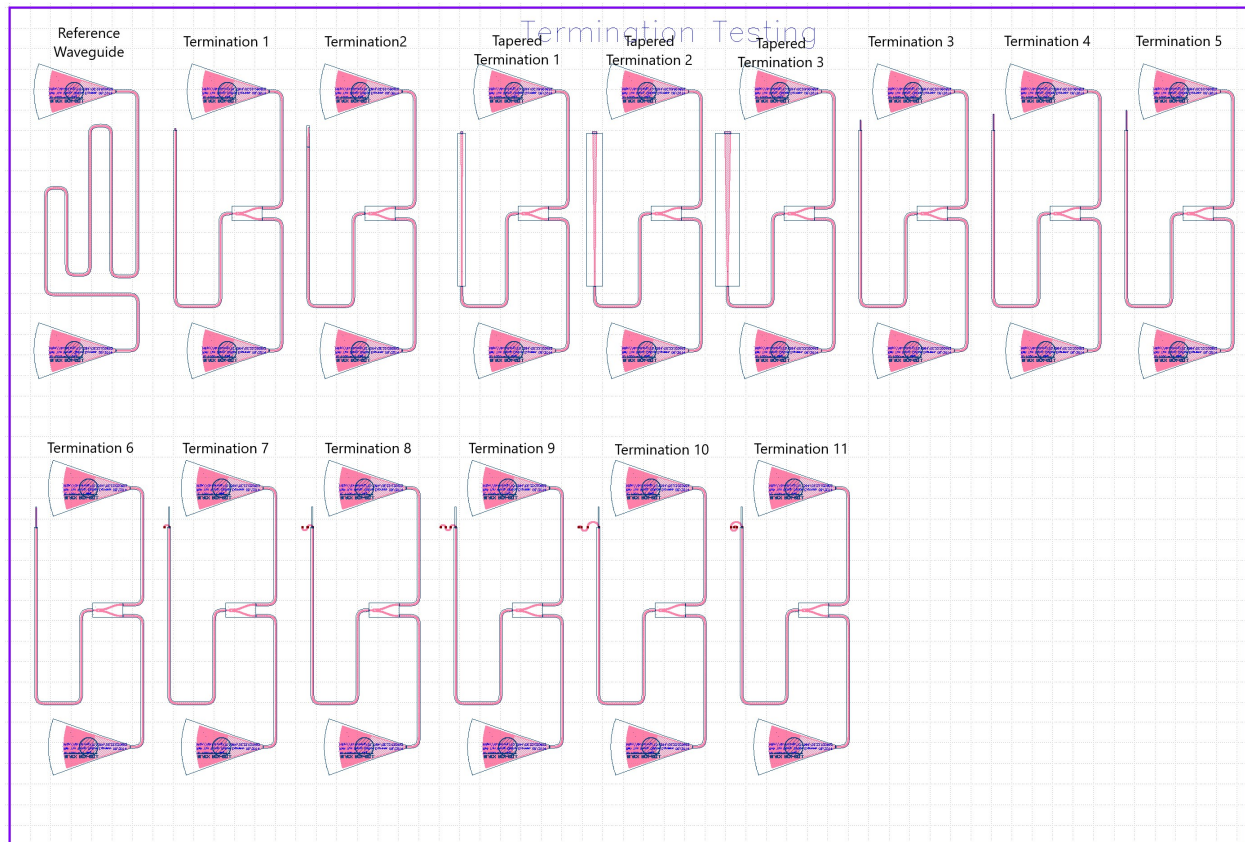
- Waveguide 1. Test station reference for a simple waveguide of approximately 750um length.
- Waveguide 2. Waveguide path length transmission comparison to understand how the wavelength length will impact the contrast from the unbalanced Mach-Zehnder Interferometers. This design assumes the y-branch is providing an ideal 50/50 split and the coupling efficiencies from the two output grating couplers to the test system are equal.
- MZI1 has 517.973 um path length imbalance, such that the fringe period will occur at close to 1.2nm spacing. This should allow fine resolution measurements of the group index.

- MZI2 has 51.65 μm path length imbalance such that the fringe period will occur at close to 12nm spacing, which results in only a few interference fringes over the 50nm coupler bandwidth.
- MZI3 is the same as MZI2, with the addition of $\pi/2$ path length change added into the long arm. I am interested to how accurately the $\pi/2$ phase shift is realized in fabrication, where MZI2 & MZI3 combined generate a pair of quadrature signals
- MZI4 is similar to MZI2, except a custom y-branch MMI has been inserted into the laser input size of the MZI. The path length imbalance is 48.932 μm for MZI4. The device was optimized by simulation in COMSOL and it will be interesting to see the real performance.
- MZI Phase 1 – 6 are unbalanced with 50 μm nominal path length, having fringe spacing similar to MZI2,3,4. The phase is shifted by reducing the trombone sections by 0.2, 0.4, 0.6, 0.8, 1, 1.2, 1.4, 1.6 λ , where λ is calculated to be 632nm for the waveguide operating with 1550nm vacuum wavelength. These structures were added to evaluate how accurate/reproducible the phase shifts can be set using an array of devices. Since there is no active phase shifter in OpenEBL, arraying the component with discrete phase shifts maybe a suitable method to evaluate phase sensitive structures.

Estimate of the FSR for the MZI are based on the analytic relationships $n_g(\lambda) = n_{eff}(\lambda) - \lambda \frac{dn_{eff}}{d\lambda}$ and $FSR(\lambda) = \frac{\lambda^2}{\Delta L n_g(\lambda)}$, where n_{eff} is the effective index for the waveguide and ΔL is the path length imbalance in the MZI.

EBeam_ChrisHamner_2ndSubmission.gds design information

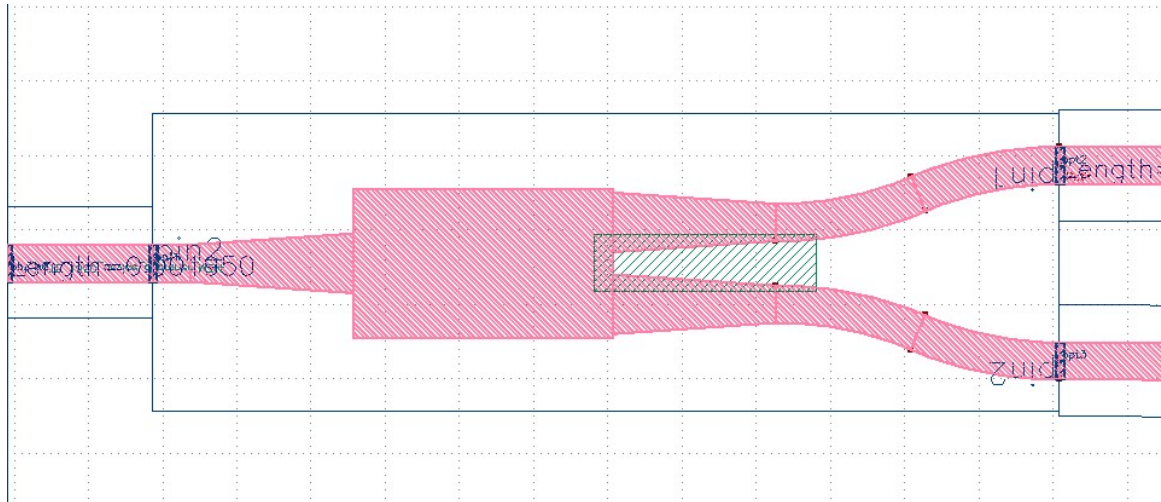
Prior to the final fabrication submission date, we were invited to submit a second design (space was available). I submitted the below 2nd design to evaluate a variety of waveguide terminations. Due to the limited time, this design was not simulated and is only being experimentally analyzed.



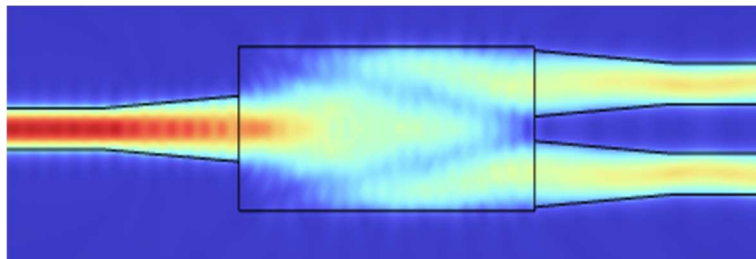
- Waveguide Reference has a path length equivalent to the double pass through the termination testing structures (without the y-branch losses).
- Termination 1 has an abrupt end to the waveguide (no termination structure).
- Termination 2 uses the available “component_ebeam_termination_te1550” taper reduced termination structure.
- Tapered Termination 1,2,3 have gradual increases in the waveguide width to 1,2,3 μm respectively, prior to an abrupt end to the waveguide.
- Termination 3,4,5,6 utilized a waveguide reduction taper combined with a sub-wavelength waveguide (transition to a sub-wavelength waveguide). With each additional design, the sub-wavelength structure is made longer (extending past the end of the taper) and the duty cycle of the sub-wavelength structure is reduced.
- Termination 7,8,9 are 180 degree circular bends with radius= $1.25\mu\text{m}$. Each additional number includes an additional 180 degree bend.
- Termination 10 uses a series of 180 degree circular bends of decreasing radius $2.75\mu\text{m}$, $2.5\mu\text{m}$, $0.75\mu\text{m}$. Termination 11 is the same, but the orientation of the bends is arranged to make a spiral.

Component Simulations:

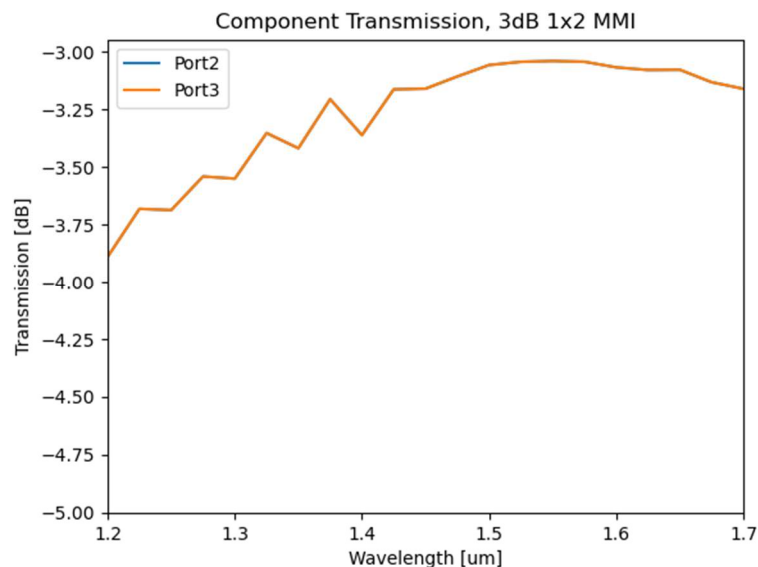
The MMI y-branch used on MZI4 is shown below. This shape optimized (waveguide tapers, MMI width & length) for maximum equal transmission.



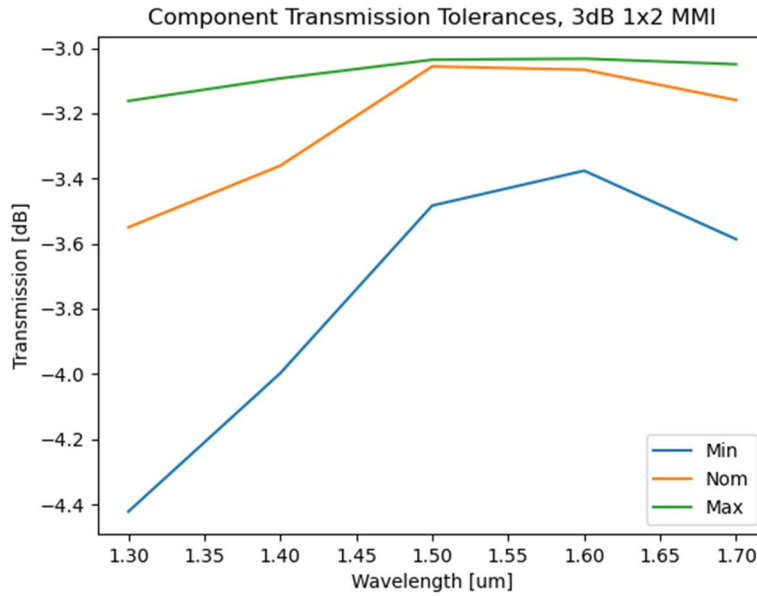
The simulation result from COMSOL is shown below at 1550nm. The design was optimized only at 1550nm while the performance was simulated from 1300 to 1700 nm.



The 1550nm 1x2 MMI was simulated for the nominal performance vs wavelength.

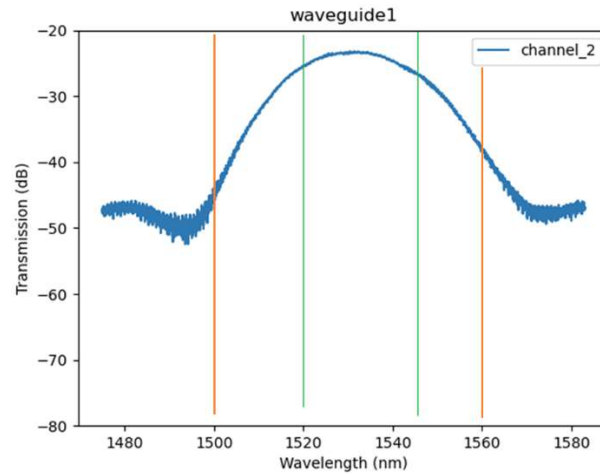


A corner simulation of the 1x2 MMI device was performed for ± 50 nm variation of the following parameters: MMI region width, MMI region length, waveguide width (all waveguides), input taper width at MMI, output taper widths at MMI, and output waveguide positions on MMI. The below plot shows the simulated performance variation.

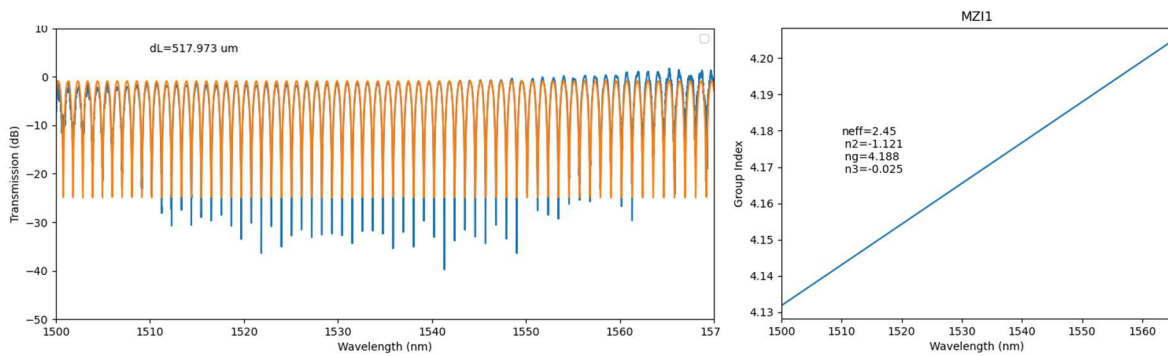


Data Analysis:

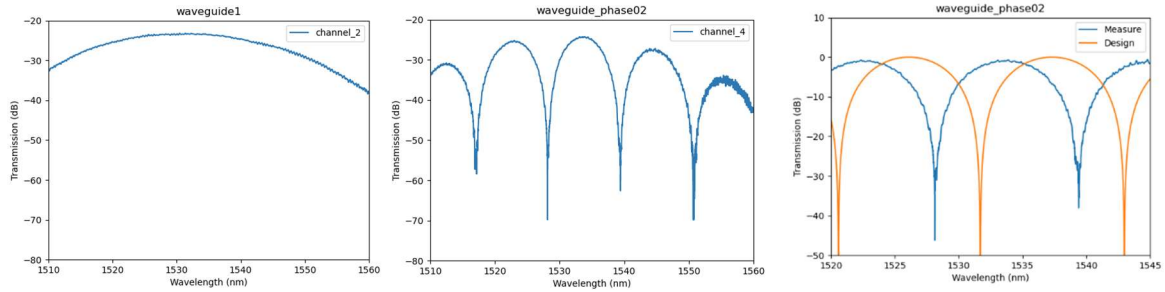
1. The measured data is truncated to the wavelength range showing larger SNR. I have chosen 1520 to 1545 for analysis of the MZI phase shift sequence, shown below. This appears to be sufficient bandwidth for the analysis and avoid the higher noise regions. Analysis of the MMI splitters and terminations will utilize a larger bandwidth of 1500nm-1560nm.



2. Group index data was extracted from MZI1 with large path length imbalance, fitting the MZI transmission function, $I(\lambda) = 10 * \log \left(\left| 1 + e^{\left(\frac{-i2\pi n_{eff} \Delta L}{\lambda} - \frac{\alpha \Delta L}{2} \right)} \right|^2 \right) + b$. The fit assumes the $\Delta L = 517.973 \text{ um}$ from the GDS design, and $n_{eff 1.55um} = 2.451$ from simulation. These values are applied for the duration of the analysis (I am assuming the index values are uniform across the test area).



- The grating coupler transmission spectra is subtracted from each data. The reference dataset is from “waveguide_1” or the “waveguide_reference”. The below plots show the reference transmission spectra (left), the measured MZI transmission spectra (middle) and the flattened transmission spectra (right).

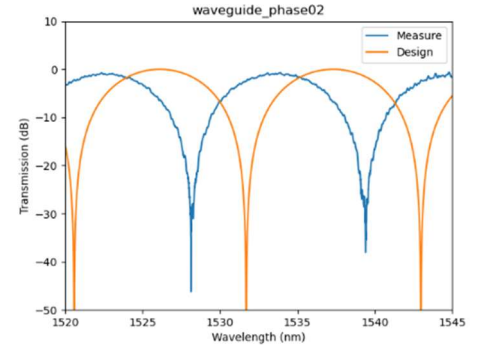


- Individual data sets are analyzed by fitting to the MZI transmission function (phase shift sequence) or making a relative comparison of transmissions (termination sequence).

Measurement analysis:

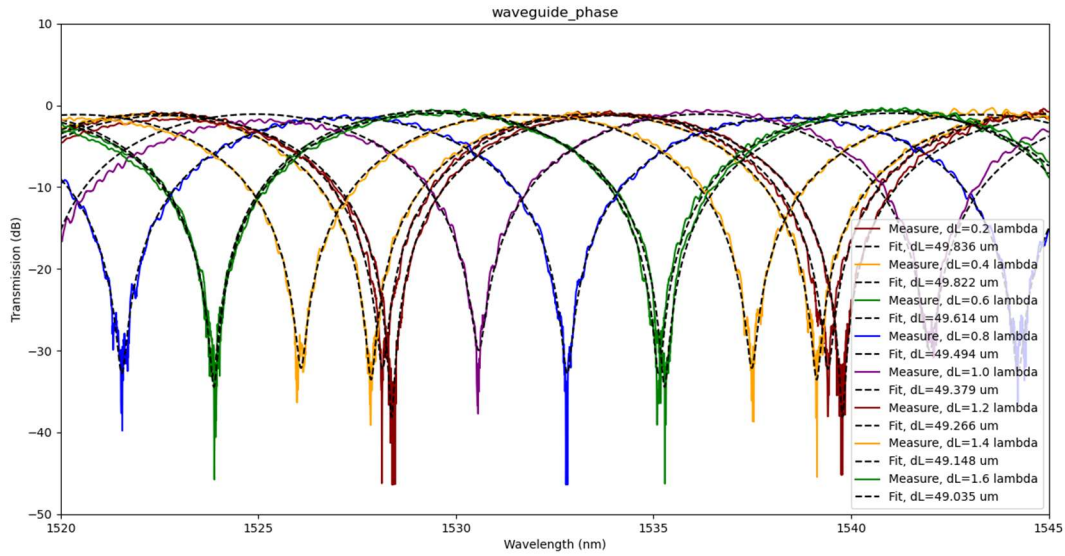
Objective: Assess the reliability of setting phase shifts generated by MZI optical path differences.

- Due to the effective index difference in simulation vs fabrication, the absolute phase difference between the simulated and measured MZI structures is approximately 0.4π . Absolute phase errors increase proportional to the path length imbalance, where these MZI's were roughly 50 μ m path length (roughly 80 waves path length).

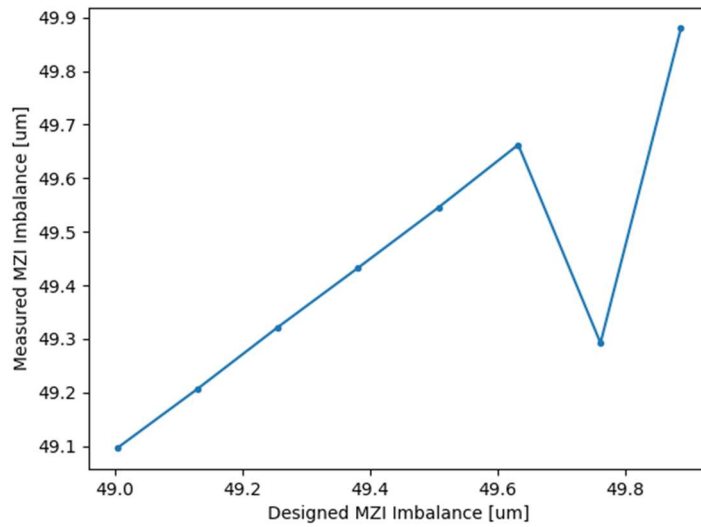


The corner simulation for n_{eff} in the TE 220 nm x 500 nm Si waveguide (215.3nm – 223.1 nm and 470 – 510 nm) results in $n_{eff\ 1.55\mu m} = 2.391 - 2.504$ and $n_g\ 1.55\mu m = 4.176 - 4.256$. The measured $n_g\ 1.55\mu m = 4.188$, such that the group index is within the anticipated range. Varying $n_{eff\ 1.55\mu m}$ over the simulated corner values has very little impact on n_g obtained from the fit data.

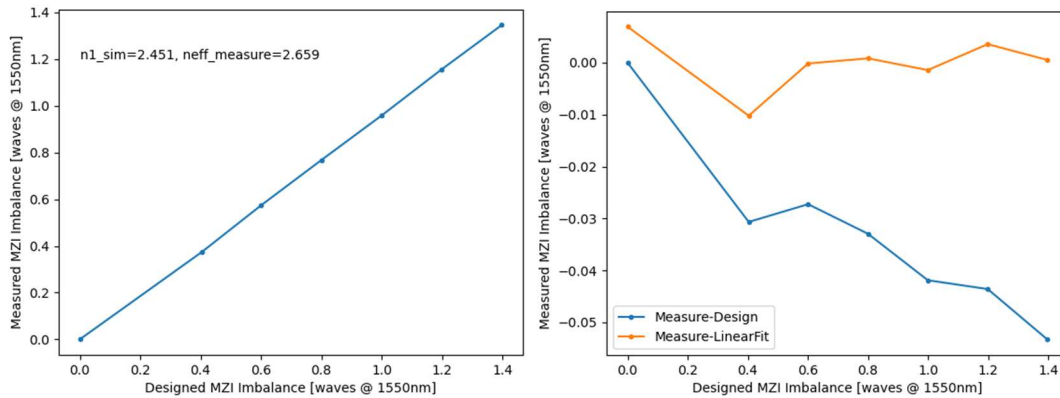
- The relative phase changes between MZIs was extracted by fitting the MZI transmission function to each, shown below.



Only 'Phase04' at 49.822 um shows significant deviation from the designed linear spacing. This data point will be omitted for the additional analysis.

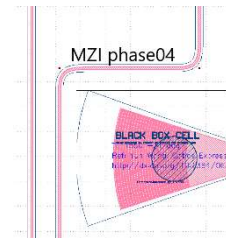


Since my objective was testing device phase sensitivity, below I recalculate the MZI output as a phase shift at 1550nm. The right plot shows the difference between measured and designed phase shifts as well as the residuals after a linear fit.



The relative phase changes between MZIs were measured to be within $1/20^{\text{th}}$ of a wave at 1550nm for phase steps up to 1.4 waves. The residual after subtracting a linear fit shows relative phase steps were within $1/100^{\text{th}}$ wave at 1550nm.

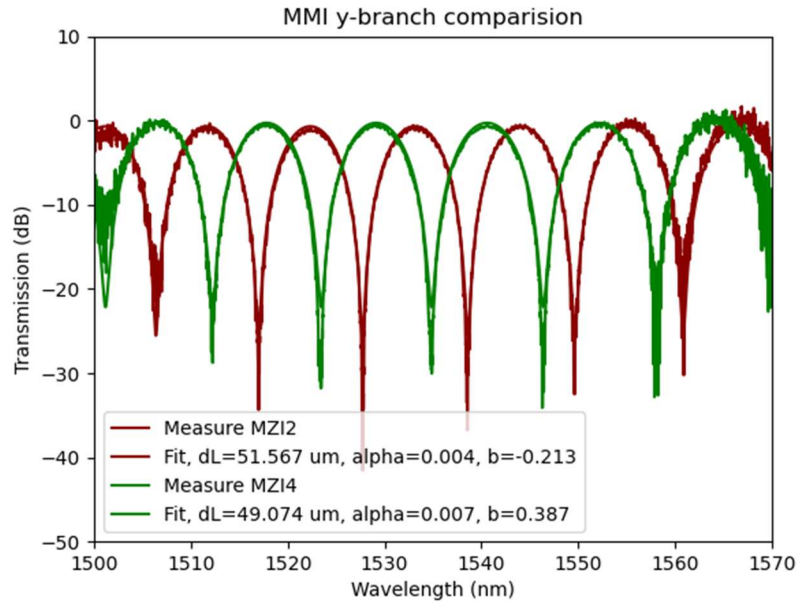
3. All but phase step “phase04” was close to the simulated result. The cause for this deviation is not clear but this is also the only MZI of this sequence that is placed in close proximity to a neighboring black box grating coupler (see right figure). Perhaps there was some interaction between these components or this MZI just has a significantly different effective index (wafer or process variation). A repeatability run would help clarify this.



While absolute phase value for the 50um unbalanced MZI was poor, the relative phase steps were close to the design values and exhibited good linearity.

Objective: Assess the relative performance of the 1x2 MMI y-branch designed in COMSOL vs the provided y-branch component.

This comparison was made by substitution of one of the provided y-branches with the MMI y-branch I designed. MZI2 is the reference case (uses 2 provided y-branch components), while MZI4 is the test case (1 MMI and 1 provided y-branch component). After flattening and fitting the MZI spectra, the results are shown below. Performance is very similar to the provide y-branch (losses and bandwidth over the tested wavelength range). The only notable difference seems to be the reduced extinction ratio for the MMI device. However, this single data point is not sufficient to determine if the difference is in the design or the fabrication tolerances. For example, the MZI phase shift sequence (previous section) also shows 10 dB variation between the extinction ratio of the nearly identical fabricated MZIs.

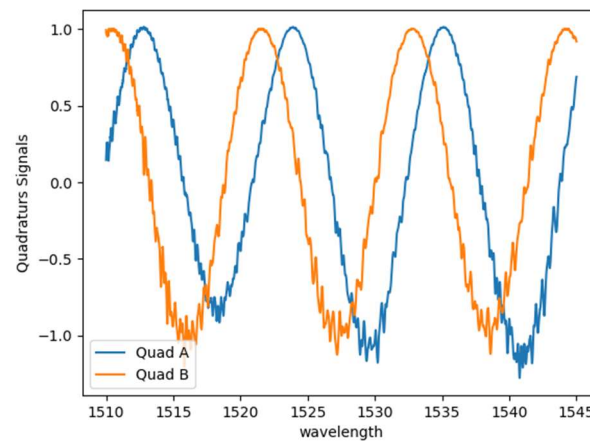


The phase shift between MZI2 and MZI4 (custom splitter) is due to the path length change required to match the different pin spacings I used for the MMI y-branch. The designed path length imbalance for MZI2, MZI4 is 51.65, 48.932 μm , while the measured values are 51.567, 49.047 μm , respectively.

This simple MMI based y-branch appears to perform well over the 1500-1560nm range, but would benefit from additional testing (repeatability & bandwidth).

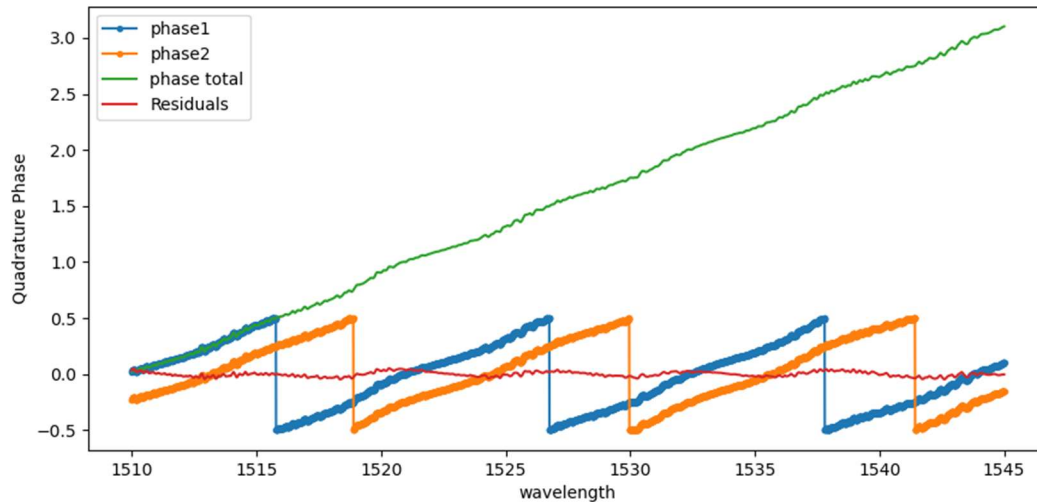
Objective: Generation of Quadrature Signals from Pair of Phase Shifted MZIs

I intended to use MZI2 & MZI4 as a pair of 0.5π shifted quadrature signals. However, the phase error was much larger for this pair of MZIs, so I am performing the analysis on Phase06 & Phase08 MZIs (0.4π phaser shift) instead.



The MZI transmission is converted to linear scale, the amplitude and offset are fit and corrected to be on the range of -1 to +1, then the arctan function is applied to extract the

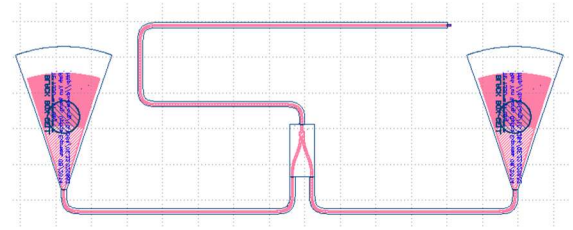
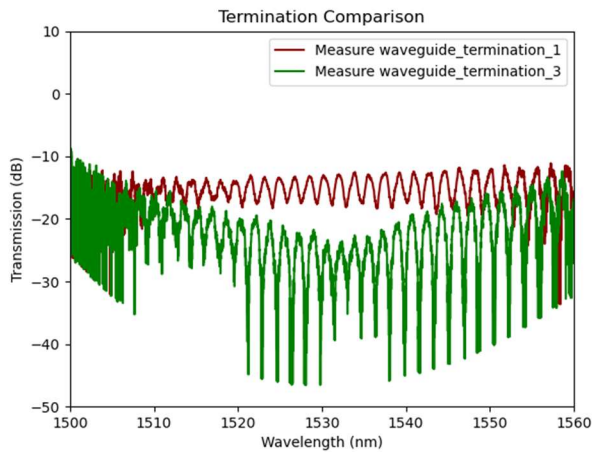
relative phases of the signal's vs wavelength. Two shifted phases can be obtained depending on the order of the quadrature signals applied to the arctan function. The result is unwrapped to generate a monotonic signal vs wavelength and residuals to a linear fit are shown below. Additional calibration would be necessary to correct the phase error, wavelength dependent amplitude, and wavelength dependent offset.



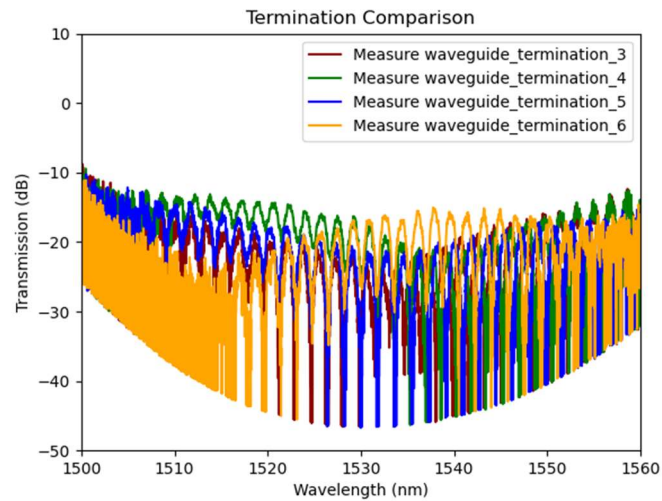
Objective: Compare Termination Types:

The second design submission was a bonus due to available space on the wafer. This set of test structures is empirically evaluating the back reflection from different waveguide termination structures. Note: Termination 2 (the PDK default termination taper) data is missing from the supplied dataset.

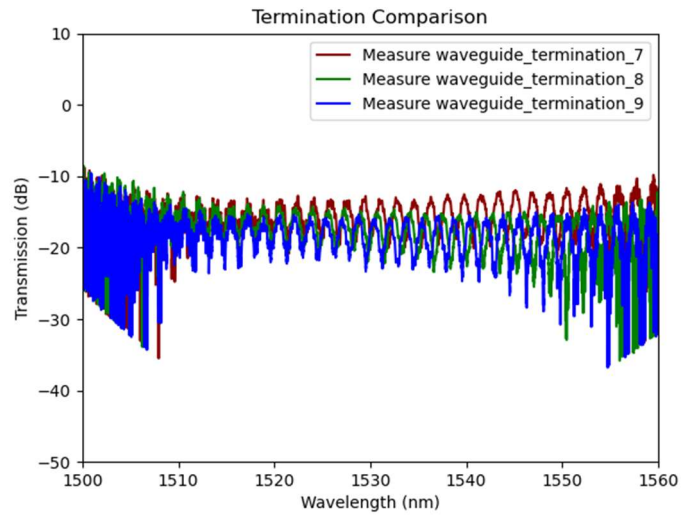
The poorest performing termination, Termination 1, was the abrupt waveguide termination (waveguide just ends), while the best was Termination 3 (which is most similar to the PDK default termination), shown in the below comparison. Note that these termination tests are double pass through a y-branch, such that the actual reflection from the terminator is 6 dB higher than measured on this test structure. As a point of comparison, Fresnel reflection from medium with $n=2.45$ to $n=1.5$ would result in roughly 6% (-12 dB reflection).



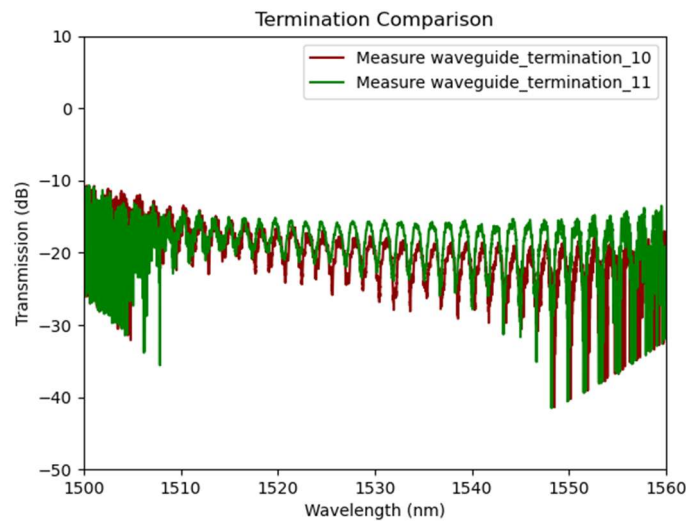
The other configurations (with intermediate or poor performance) are grouped into their design categories. Terminations 3,4,5,6 were varying length reduced tapers including increasingly long sub-wavelength sections.



Terminations 7,8,9 were increasing number (1,2,3) of 180 degree, radius=1.25um waveguide bends.



Terminations 10,11 were 3 waveguide bends of decreasing radius.



Applied Nanotools, Inc. NanoSOI process:

The photonic devices were fabricated using the NanoSOI MPW fabrication process by Applied Nanotools Inc. (<http://www.appliednt.com/nanosoi>; Edmonton, Canada) which is based on direct-write 100 keV electron beam lithography technology. Silicon-on-insulator wafers of 200 mm diameter, 220 nm device thickness and 2 μm buffer oxide thickness are used as the base material for the fabrication. The wafer was pre-diced into square substrates with dimensions of 25x25 mm, and lines were scribed into the substrate backsides to facilitate easy separation into smaller chips once fabrication was complete.

After an initial wafer clean using piranha solution (3:1 H₂SO₄:H₂O₂) for 15 minutes and water/IPA rinse, hydrogen silsesquioxane (HSQ) resist was spin-coated onto the substrate and heated to evaporate the solvent. The photonic devices were patterned using a JEOL JBX-8100FS electron beam instrument at The University of British Columbia. The exposure dosage of the design was corrected for proximity effects that result from the backscatter of electrons from exposure of nearby features. Shape writing order was optimized for efficient patterning and minimal beam drift. After the e-beam exposure and subsequent development with a tetramethylammonium sulfate (TMAH) solution, the devices were inspected optically for residues and/or defects. The chips were then mounted on a 4" handle wafer and underwent an anisotropic ICP-RIE etch process using chlorine after qualification of the etch rate. The resist was removed from the surface of the devices using a 10:1 buffer oxide wet etch, and the devices were inspected using a scanning electron microscope (SEM) to verify patterning and etch quality. A 2.2 µm oxide cladding was deposited using a plasma-enhanced chemical vapour deposition (PECVD) process based on tetraethyl orthosilicate (TEOS) at 300°C. Reflectometry measurements were performed throughout the process to verify the device layer, buffer oxide and cladding thicknesses before delivery.

Measurement description:

To characterize the devices, a custom-built automated test setup [2, 6] with automated control software written in Python was used [3]. An Agilent 81600B tunable laser was used as the input source and Agilent 81635A optical power sensors as the output detectors. The wavelength was swept from 1500 to 1600 nm in 10 pm steps. A polarization maintaining (PM) fibre was used to maintain the polarization state of the light, to couple the TE polarization into the grating couplers [4]. A 90° rotation was used to inject light into the TM grating couplers [4]. A polarization maintaining fibre array was used to couple light in/out of the chip [5].

Acknowledgments:

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References:

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- [2] Lukas Chrostowski, Michael Hochberg, chapter 12 in "Silicon Photonics Design: From Devices to Systems", Cambridge University Press, 2015
- [3] <http://siepic.ubc.ca/probestation>, using Python code developed by Michael Caverley.
- [4] Yun Wang, Xu Wang, Jonas Flueckiger, Han Yun, Wei Shi, Richard Bojko, Nicolas A. F. Jaeger, Lukas Chrostowski, "Focusing sub-wavelength grating couplers with low back reflections for rapid prototyping of silicon photonic circuits", Optics Express Vol. 22, Issue 17, pp. 20652-20662 (2014) doi: 10.1364/OE.22.020652
- [5] www.plcconnections.com, PLC Connections, Columbus OH, USA.
- [6] <http://mapleleafphotonics.com>, Maple Leaf Photonics, Seattle WA, USA.